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Lab 4.2 MTConnect Practice

2.1 Run MTConnect Agent on Laptop

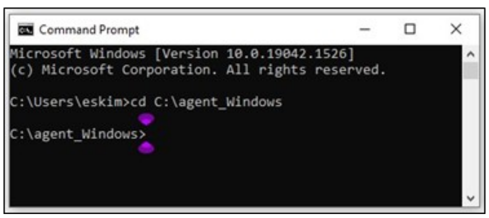
Now, we are ready to run MTConnect agent. First, we will run MTConnect agent on laptop without change from the given configurations. Please follow the direction below.

1. Open 'Command Prompt'.
2. Change directory by 'cd' and <file directory> command as below:

Windows - Command Prompt

```
cd <agent file directory>
```

- In the example below, the file directory is C:\agent_Windows



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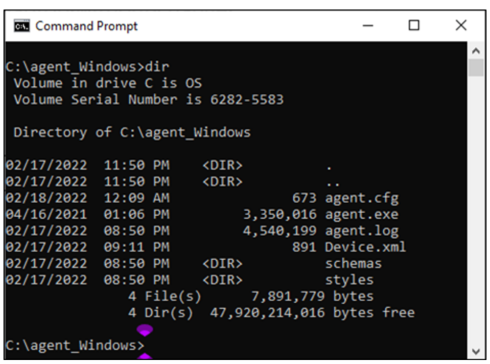
3. (Optional) To see the contents in the directory, use 'dir' command.



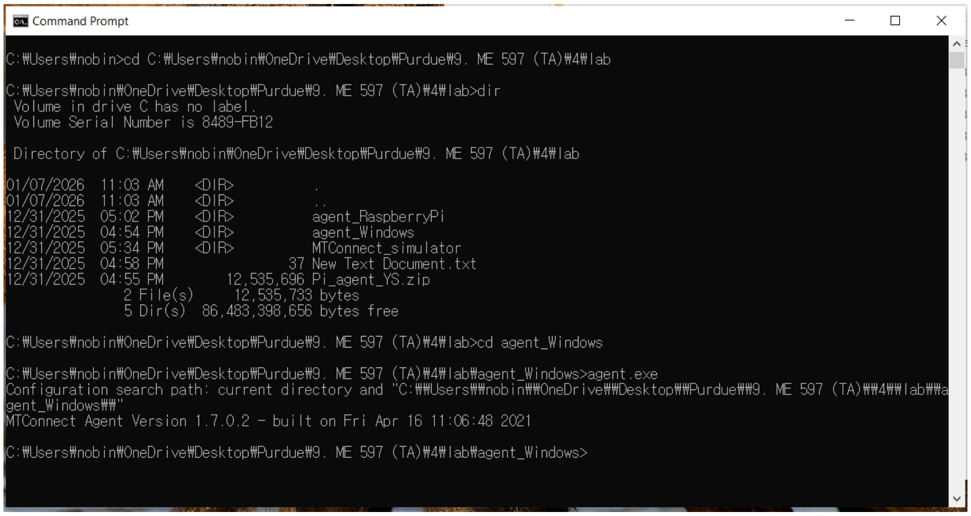
4. Run MTConnect agent by 'agent.exe run' command.

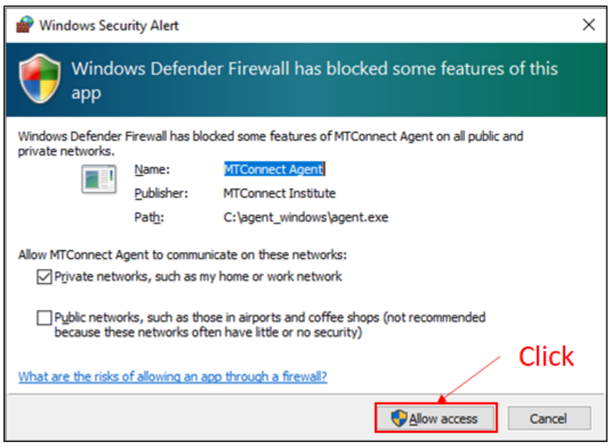
a. If 'Windows Security Alert' pops up, click 'Allow access'.

Windows - Command Prompt

```
agent.exe run
```

- If you want to run agent as a debug mode, use agent.exe debug.





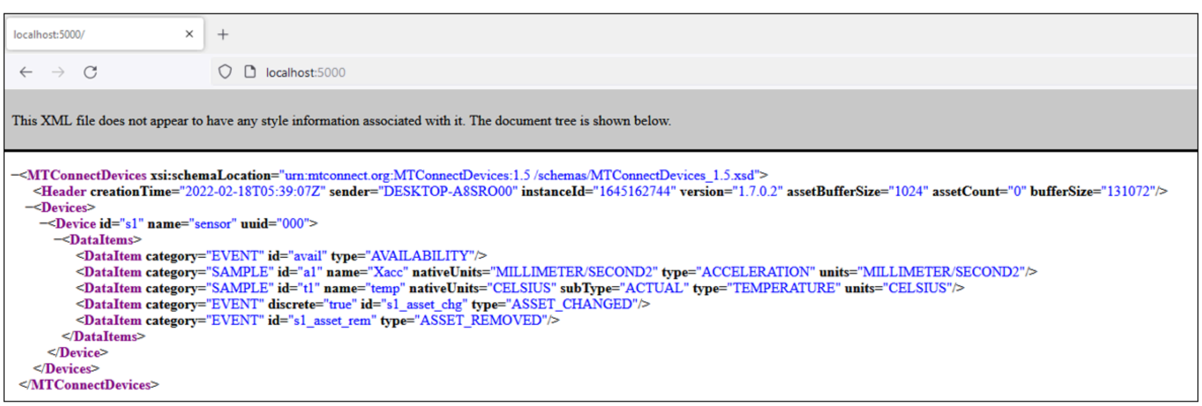
5. To check MTConnect agent that you run, open a web-browser on laptop, and then type ‘localhost:5000’ on the address bar.

a. If you see the capture below, you succeed to run MTConnect agent!

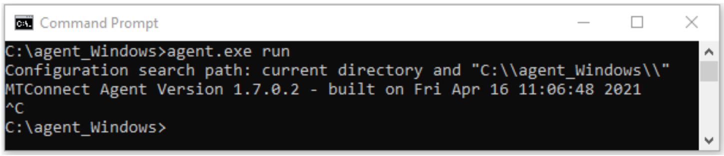
b. Other IP address should work such as ‘127.0.0.1:5000’ and ‘your IP address:5000’

c. This request provides MTConnect properties that is defined in ‘Device.xml’.

d. Try to compare ‘Device.xml’ and the request.



6. Try 3 different data collection methods below.
7. To close (halt) the agent service, press Ctrl + c on ‘Command Prompt’.



Tip - Data Collection Method

Data collection methods may vary depending on the client applications' needs but a general method of collecting data from an MTConnect agent is described below. The method described is used to aggregate 'ALL' data from an MTConnect agent and ensure data is not duplicated in a data store and not missed in the agent buffer. All responses upon HTTP requests are XML documents.

Generally, there are 3 steps to collecting data from MTConnect.

1. Map the properties (Probe the agent)

a. IP_Address:Port_Number/probe

```
i.      Ex) http://127.0.0.1:5000/probe
```

b. Probe request provides device data model (Device.xml) – including dataitems and metadata 2. Request 'Current' state of machine

a. IP_Address:Port_Number/current

```
i.      Ex) http://127.0.0.1:5000/current
```

b. Current request provides current state (the latest) of 'ALL' dataitems 3. Periodically get 'ALL' data from the agent from where you last left off and a reasonably large 'count' to ensure you get everything in the agent (you normally base this on the sample rate of the adapter) a. IP_Address:Port_Number/sample

```
i. Ex) http://127.0.0.1:5000/sample
```

b. Sample request provides 100 samples (default) from the first sequence.

c. In addition, you'd like to request more samples and specify starting sequence, refer to the example below.

```
i.      Ex) http://127.0.0.1:5000/sample?from=3&count=200

ii.     This request getting the next 200 samples of data published to the agent from
sequence number 3.
```

Task 2.1

1. Change port number in *'agent.cfg'* from 5000 to 5001 of the agent and then run on laptop.
2. Request *'127.0.0.1:5001/current'* on a web browser of **laptop** and then capture and attach it below. If you get a **blank screen**, please see the following '*Blank screen solution*'.
3. Request *'IP_Address:5001/current'* on a web browser (*'Chromium'*) of **Raspberry Pi** and then capture and attach it below. If you get any error, please try to turn off your fire walls on your laptop.

Blank screen solution: You may get a blank screen that nothing is shown in your browser.

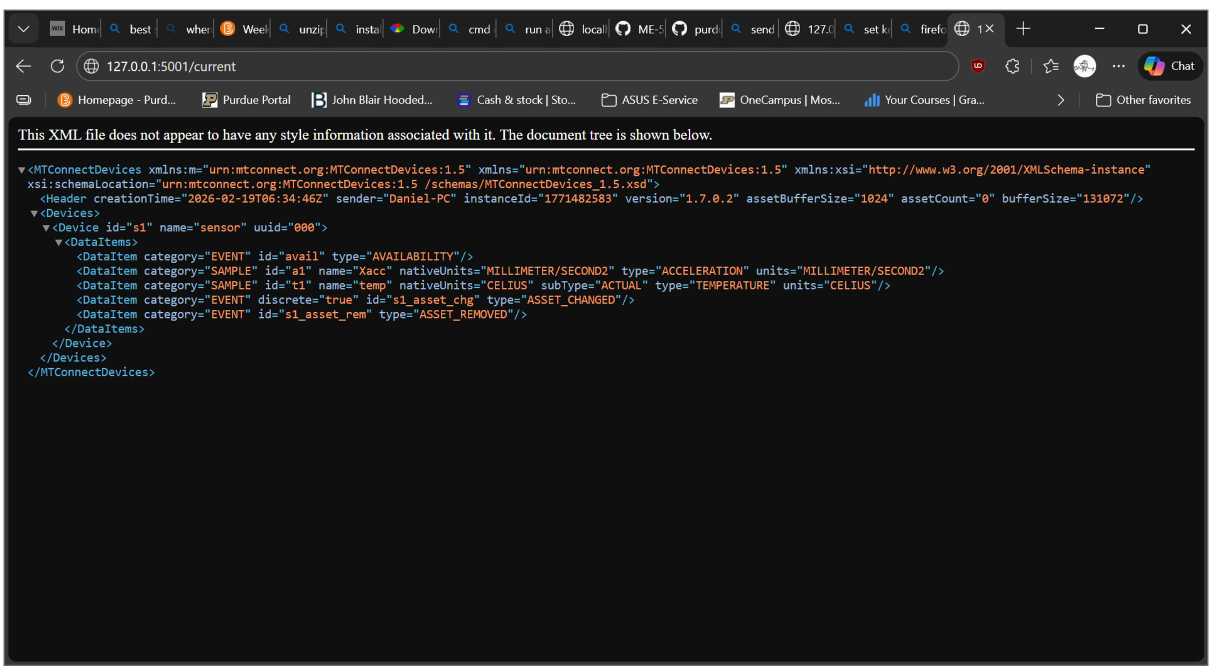
- Method 1 : Press 'Ctrl+U' to see the page source of the URL.
- Method 2 :

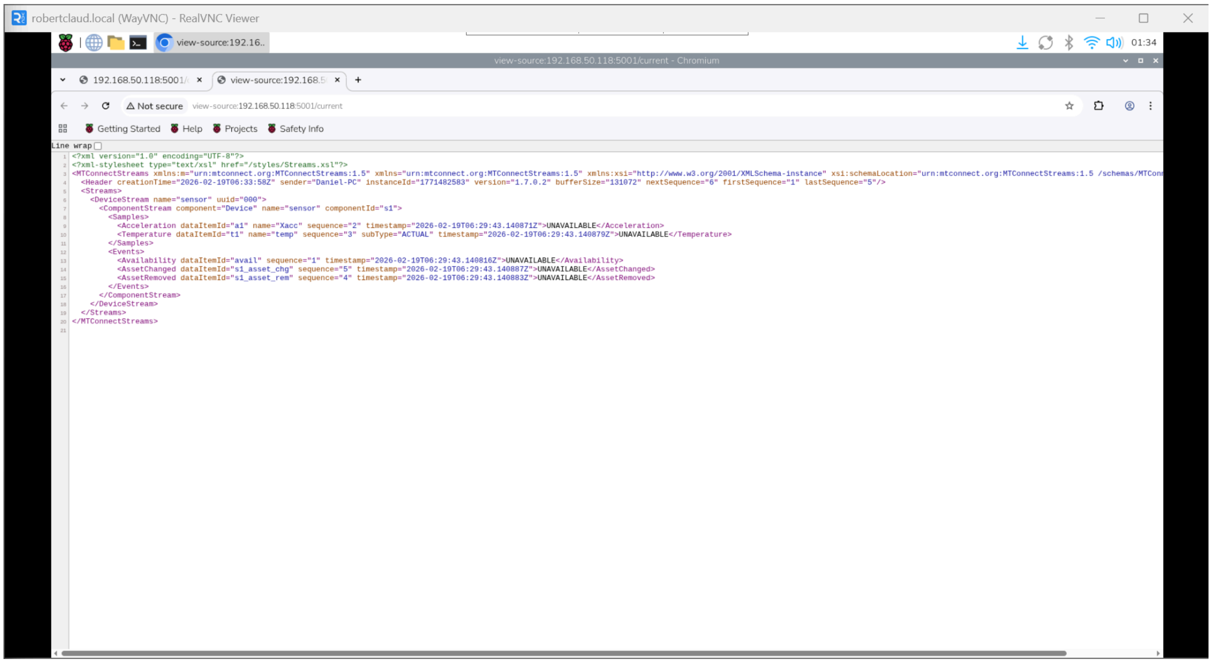
1. Open your '*agent.cfg*' file using a text editor and do the following changes as shown in the following figure:

2. In line 7, change '*Pretty = true*' to '*Pretty = false*'.
3. Comment out the contents between line 18 (included) and line 35 (included). Your updated '*agent.cfg*' file will look like the following figure.

```
1  Devices = ./Device.xml
2  Port = 5000
3  AllowPut = true
4  ReconnectInterval = 1000
5  BufferSize = 17
6  MonitorConfigFiles = true
7  Pretty = false # true
8  SchemaVersion = 1.5
9
10 Adapters {
11    # Log file has all machines with device name prefixed
12    Adapter1 {
13       Host = IP Address
14       Port = Port Number
15    }
16 }
17
18 # Files {
19 #     schemas {
20 #         Path = ./schemas
21 #         Location = /schemas/
22 #     }
23 #     styles {
24 #         Path = ./styles
25 #         Location = /styles/
26 #     }
27 #     Favicon {
28 #         Path = ./styles/favicon.ico
29 #         Location = /favicon.ico
30 #     }
31 # }
32 #
33 # StreamsStyle {
34 #   Location = ./styles/Streams.xsl
35 # }
36
37 # Logger Configuration
38 logger_config
39 {
40     logging_level = debug
41     output = cout
42 }
43
```

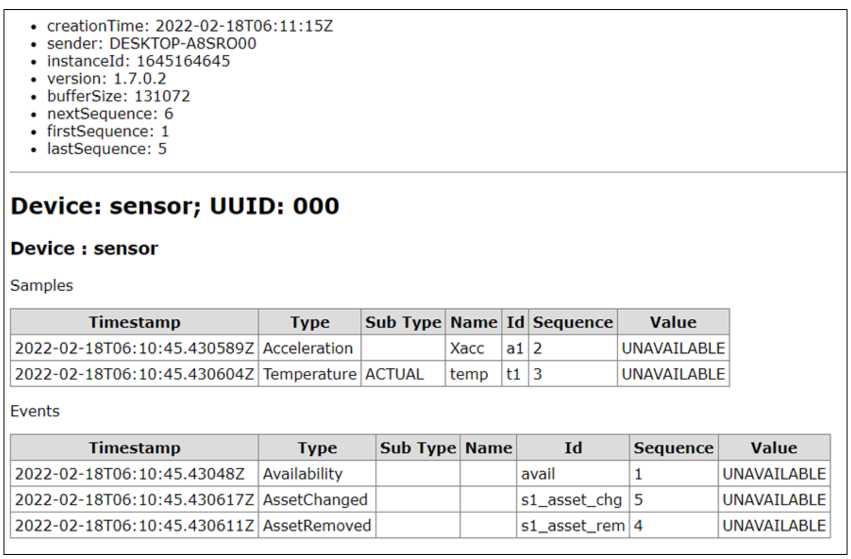
After making this change, you will see the XML file instead of the 'Pretty' style. However, both of them have the same information. If you cannot see the 'Pretty' style, you will use the XML sytle in the following labs. For the difference between the 'Pretty' style and the XML style, please see Figure 5 and Figure 6.





Tip - Change view of MTConnect agent data

When you request using 'current' or 'sample' method, the response looks as in Figure 5. This is because of the styles how to show XML document. If you want to see data in XML document format as Figure 6, right click on the web browser and then click 'View page source'.



- creationTime: 2022-02-18T06:11:15Z
- sender: DESKTOP-A8SRO00
- instanceId: 1645164645
- version: 1.7.0.2
- bufferSize: 131072
- nextSequence: 6
- firstSequence: 1
- lastSequence: 5

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-02-18T06:10:45.430589Z	Acceleration		Xacc	a1	2	UNAVAILABLE
2022-02-18T06:10:45.430604Z	Temperature	ACTUAL	temp	t1	3	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2022-02-18T06:10:45.43048Z	Availability			avail	1	UNAVAILABLE
2022-02-18T06:10:45.430617Z	AssetChanged			s1_asset_chg	5	UNAVAILABLE
2022-02-18T06:10:45.430611Z	AssetRemoved			s1_asset_rem	4	UNAVAILABLE

Figure 5 Default view of current data from agent

```xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="./styles/Streams.xsl"?>
<MTConnectStreams xmlns:m="urn:mtconnect.org:MTConnectStreams:1.5" xmlns="urn:mtconnect.org:MTConnectStreams:1.5"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="urn:mtconnect.org:MTConnectStreams:1.5
/schemas/MTConnectStreams_1.5.xsd">
  <Header creationTime="2022-02-18T06:11:15Z" sender="DESKTOP-A8SRO00" instanceId="1645164645" version="1.7.0.2" bufferSize="131072"
nextSequence="6" firstSequence="1" lastSequence="5"/>
  <Streams>
    <DeviceStream name="sensor" uuid="000">
      <ComponentStream component="Device" name="sensor" componentId="s1">
        <Samples>
          <Acceleration dataItemId="a1" name="Xacc" sequence="2" timestamp="2022-02-18T06:10:45.430589Z">UNAVAILABLE</Acceleration>
          <Temperature dataItemId="t1" name="temp" sequence="3" subType="ACTUAL" timestamp="2022-02-18T06:10:45.430604Z">UNAVAILABLE</Temperature>
        </Samples>
        <Events>
          <Availability dataItemId="avail" sequence="1" timestamp="2022-02-18T06:10:45.43048Z">UNAVAILABLE</Availability>
          <AssetChanged dataItemId="s1_asset_chg" sequence="5" timestamp="2022-02-18T06:10:45.430617Z">UNAVAILABLE</AssetChanged>
          <AssetRemoved dataItemId="s1_asset_rem" sequence="4" timestamp="2022-02-18T06:10:45.430611Z">UNAVAILABLE</AssetRemoved>
        </Events>
      </ComponentStream>
    </DeviceStream>
  </Streams>
</MTConnectStreams>
```


Figure 6 XML document view of current data from agent

2.2 Run MTConnect agent on Raspberry Pi

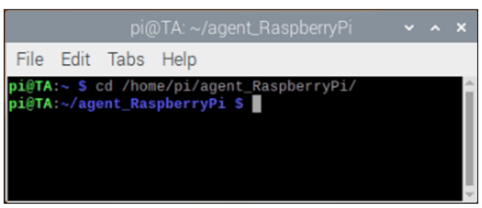
Let's run MTConnect agent on Raspberry Pi. Please follow the steps below.

1. Open 'Terminal'.
2. Change directory using 'cd' <file directory> command as below.

Raspberry Pi - Terminal

```
cd /home/pi/agent_RaspberryPi/
```

- In this example, the file directory is /home/pi/agent_RaspberryPi/.



- You must give execute permission before run program files.

Raspberry Pi - Terminal

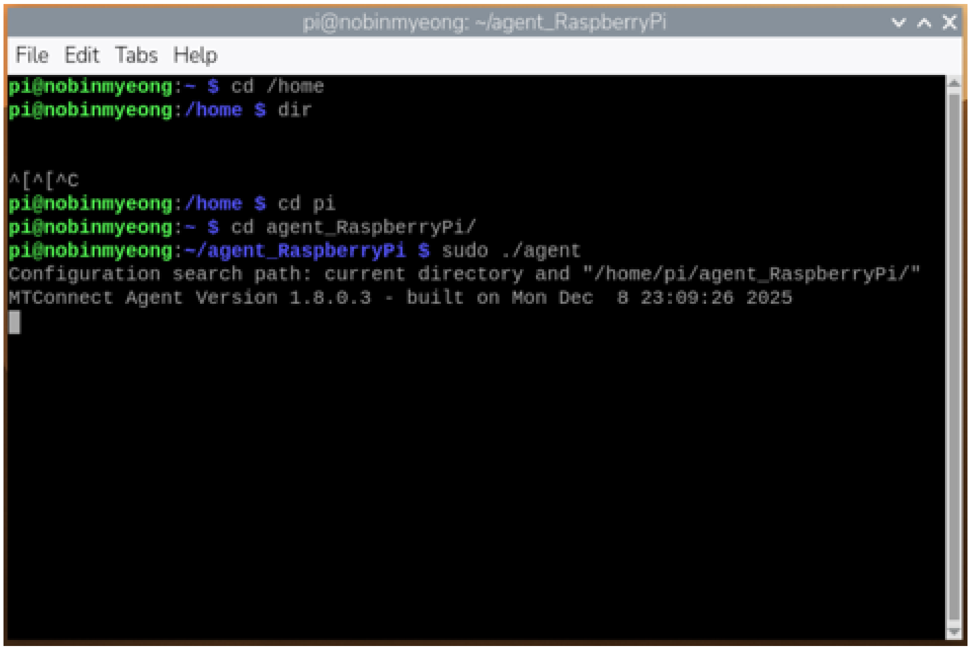
```
sudo chmod 777 *
```

---

3. Run MTConnect agent by 'sudo ./agent' command.

Raspberry Pi - Terminal

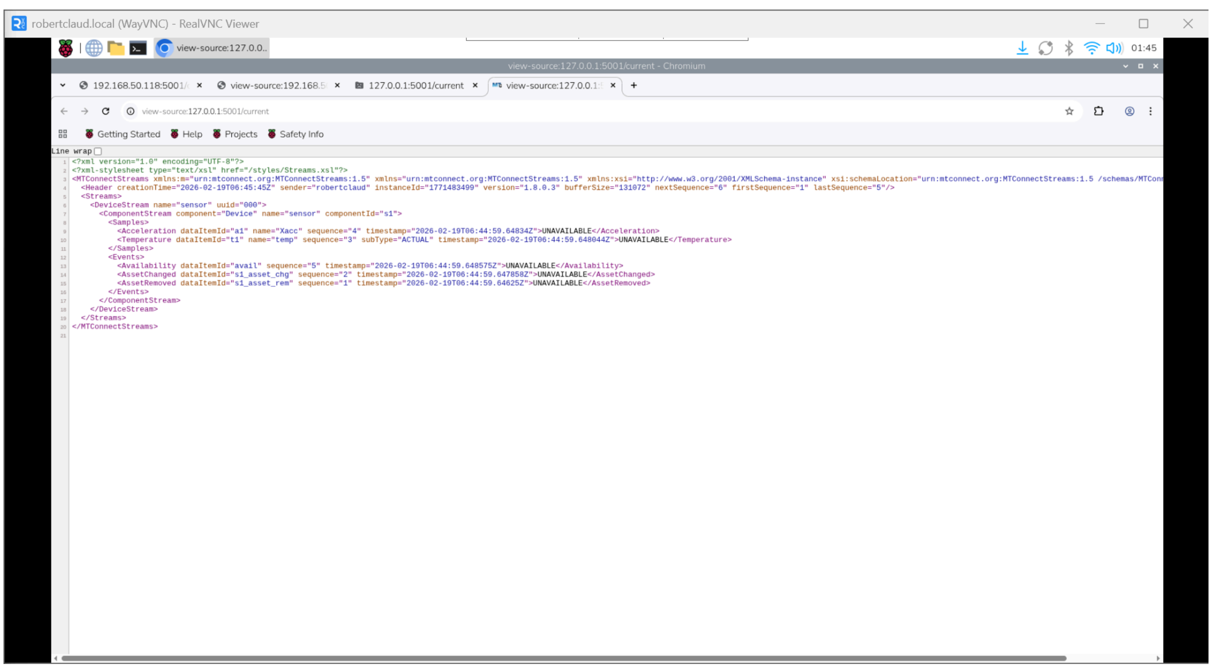
```
sudo ./agent
```

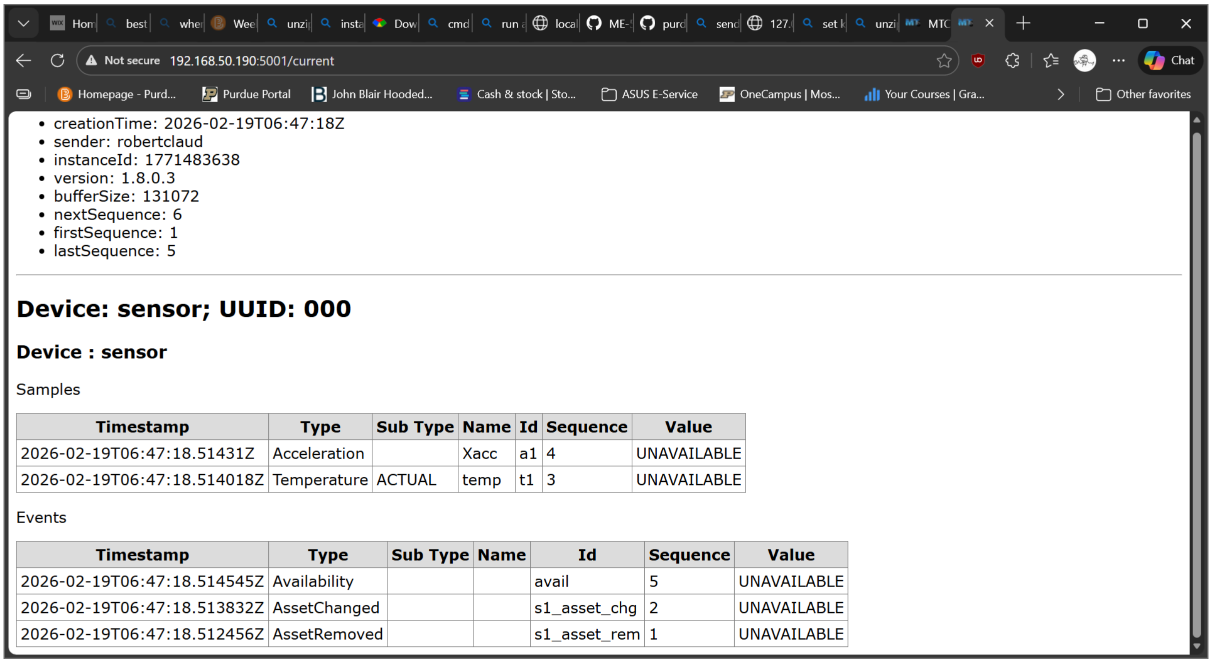


4. To check MTConnect agent that you run, use a web browser.
5. To close (halt) the agent service, press Ctrl + c on *'Terminal'*.

Task 2.2

1. Change port number in 'agent.cfg' from 5000 to 5001 of the agent and then run on **Raspberry Pi**.
2. Request *'127.0.0.1:5001/current'* on a web browser of **Raspberry Pi** and then capture and attach it below.
3. Request *'Pi_IP_Address:5001/current'* on a web browser of **laptop** and then capture and attach it below.





192.168.50.190:5001/current

- creationTime: 2026-02-19T06:47:18Z
- sender: robertclaud
- instanceId: 1771483638
- version: 1.8.0.3
- bufferSize: 131072
- nextSequence: 6
- firstSequence: 1
- lastSequence: 5

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-02-19T06:47:18.51431Z	Acceleration		Xacc	a1	4	UNAVAILABLE
2026-02-19T06:47:18.514018Z	Temperature	ACTUAL	temp	t1	3	UNAVAILABLE

Events

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-02-19T06:47:18.514545Z	Availability			avail	5	UNAVAILABLE
2026-02-19T06:47:18.513832Z	AssetChanged			s1_asset_chg	2	UNAVAILABLE
2026-02-19T06:47:18.512456Z	AssetRemoved			s1_asset_rem	1	UNAVAILABLE

2.3 Modify device data model

In this part, we will change the device data model by modifying 'Device.xml'. The given device model has 2 data items (id="a1" and id="t1") because id="avail" is a default EVENT data item for a device. The scenario is to make MTConnect agent (laptop) device data model to collect 3-axis accelerometer (ADXL345, Lab2) data and temperature sensor (DS18B20, Lab1) data from Raspberry Pi. Additionally, let's assume there are virtual sensor which detect humidity. Because the data items are measured values, the category must be SAMPLE. The device data model schematic is shown in Figure 7. Therefore, your Raspberry Pi will be an MTConnect adapter and your laptop will be an MTConnect agent. Please follow the steps below.

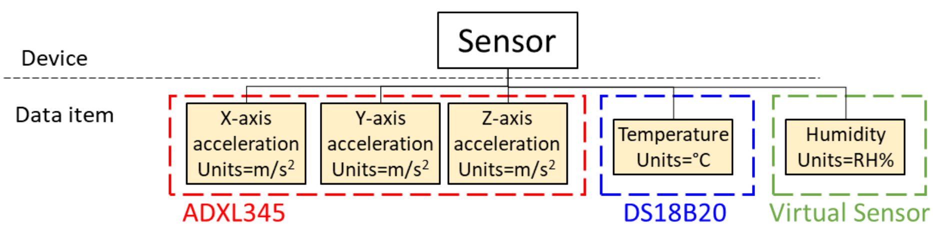


Figure 7 Device data model schematic

1. Open 'Device.xml' using any text editors on laptop.

```
a.      You can use any kind of text editors, but Notepad++ ([download here]
(https://notepad-plus-plus.org/downloads/)) is recommended.
```

2. Add 3 more 'DeviceItem' elements between Line 8 and 9 as below.

```
a.      \<DataItem category="SAMPLE" id="a2" name="Yacc" type="ACCELERATION"
units="MILLIMETER/SECOND2"/>
```

```
b.      \<DataItem category="SAMPLE" id="a3" name="Zacc" type="ACCELERATION"
units="MILLIMETER/SECOND2"/>
```

```
c.      \<DataItem category="SAMPLE" id="h1" name="humd" type="HUMIDITY_RELATIVE"
subType="ACTUAL" units="PERCENT"/>
```

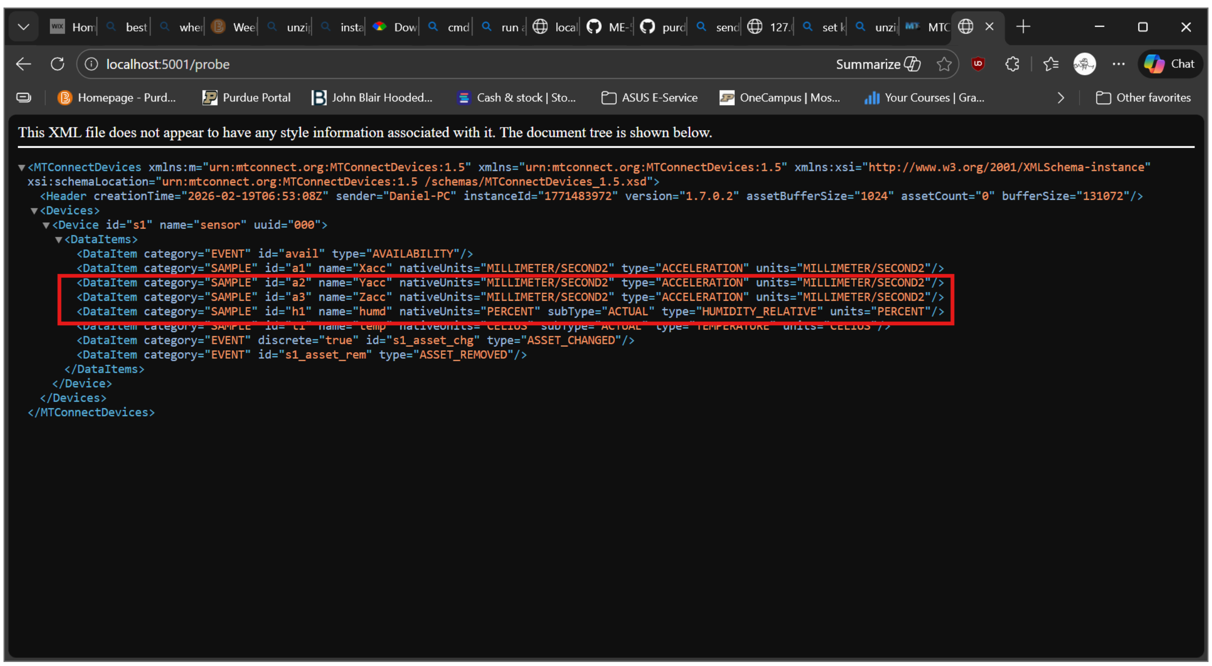
```xml
<?xml version="1.0" encoding="UTF-8"?>
<MTConnectDevices xmlns:m="urn:mtconnect.org:MTConnectDevices:1.1" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns=
"urn:mtconnect.org:MTConnectDevices:1.1" xsi:schemaLocation="urn:mtconnect.org:MTConnectDevices:1.1
http://www.mtconnect.org/schemas/MTConnectDevices_1.1.xsd">
    <Header creationTime="2010-03-04T18:44:40+00:00" sender="localhost" instanceId="1267728234" bufferSize="131072" version=
    "1.1"/>
  <Devices>
    <Device id="dev" name="default" uuid="000">
      <DataItems>
        <DataItem category="EVENT" id="avail" type="AVAILABILITY"/>
        <DataItem category="SAMPLE" id="a1" name="Xacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
        <DataItem category="SAMPLE" id="a2" name="Yacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
        <DataItem category="SAMPLE" id="a3" name="Zacc" type="ACCELERATION" units="MILLIMETER/SECOND2"/>
        <DataItem category="SAMPLE" id="h1" name="humd" type="HUMIDITY_RELATIVE" subType="ACTUAL" units="PERCENT"/>
        <DataItem category="SAMPLE" id="t1" name="temp" type="TEMPERATURE" subType="ACTUAL" units="CELSIUS"/>
      </DataItems>
    </Device>
  </Devices>
</MTConnectDevices>
```

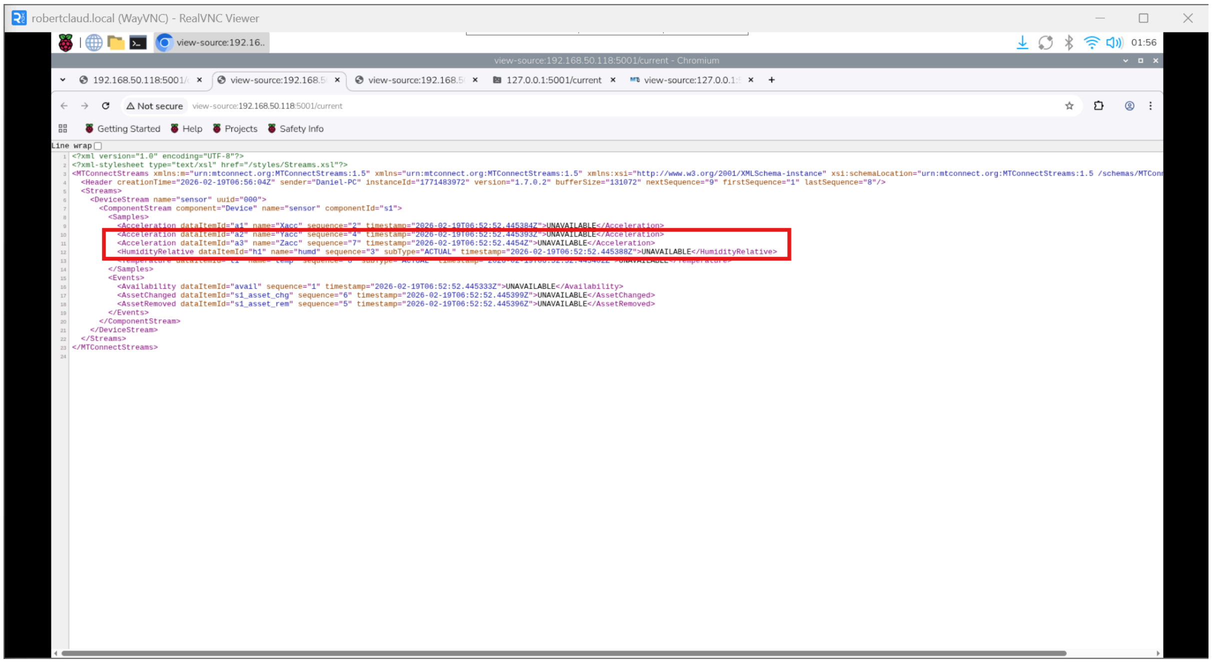
3. Save the file and then close

Task 2.3

1. Run the agent on laptop after modifying 'Device.xml' as Sec2.3 (above).
2. Request ‘probe’ data on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
3. Request ‘current’ data on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
4. Confirm the added data items from the agent data by indicating on the captures.

- Because there are no adapters, all values are ‘UNAVAILABLE’ now.





Please continue to Lab 4.3 here.